Assumptions, dummies, logs and curves

International Marketing Analytics

Dr Arne Floh, Senior Lecturer in Marketing
WU Vienna, Institute for International Business

arne.floh@wu.ac.at
Office hours: book online (Google Calendar)

DECK 01E · WT 2026/27

Today

Assump-tions | Dummy coding | Log-log | Non-linear | Python | In class

Theory, Parts 1 to 4

- Six assumptions: look, test, fix
- Dummies for categories, the reference group
- Log-log and the other log forms
- Curves: squares, logs, the turning point

Practice, Parts 5 and 6

- statsmodels tests, polars dummies, plotnine plots
- Online quiz 3 and its answers
- Two coding cases: chocolate and fashion
- The same tasks with Posit Assistant

Readings: Skiera, Reiner & Albers (2022), Regression Analysis; Bojinov, Parzen & Hamilton (2025), Linear Regression (HBS).

Where we are: two models for brand 1

```
linear:  sales    = b0 + b1·price1 + b2·feature1 + b3·display1 + b4·temp + b5·december
log-log: ln sales = b0 + b1·ln price1 + b2·ln price2 + b3·ln price3 + b4·ln price4
                  + b5·feature1 + b6·display1 + b7·temp + b8·december
```

	Linear	Log-log
Own price	−678 units per EUR	−3.68 (elasticity)
R^2	0.896	0.903
Outcome	sales (units)	log sales

Same data, two models: which one may we trust? R^2 cannot tell, because the outcomes differ. The residuals can.

Sources: make_assumptions_figures.py on data/legacy/chocolate_dataset.csv; deck 01b, Parts 6 to 9

Assumptions and how to test them

Part 1 of 6 · look first, then test, then fix

Six assumptions, three steps each

Assumption	Look	Test (H0: assumption holds)	Fix
1 Linear form	residuals vs fitted: curve?	RESET	logs, squares
2 Constant variance	residuals vs fitted: funnel?	Breusch-Pagan	logs; HC3 errors
3 Normal residuals	Q-Q plot, histogram	Jarque-Bera, Shapiro-Wilk	logs; large n helps
4 Independent errors	residuals over time	Durbin-Watson ≈ 2	HAC errors; Session 4
5 No multicollinearity	correlation matrix	VIF below 5 (10)	drop, combine, more data
6 No dominant week	Cook's distance by week	above 4/n: inspect	check data, with and without

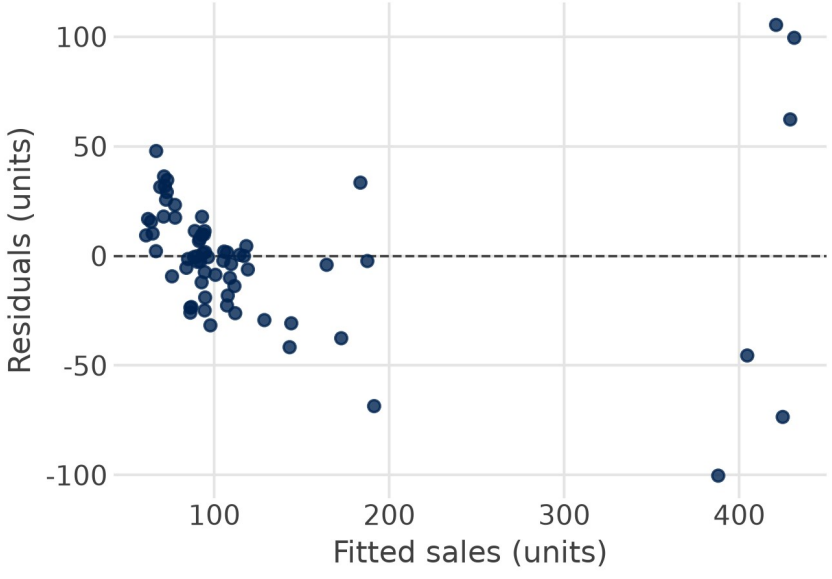
Sources: Skiera, Reiner & Albers 2022; Wooldridge 2025, ch. 8 to 9; Hair et al. 2019

How to read an assumption test

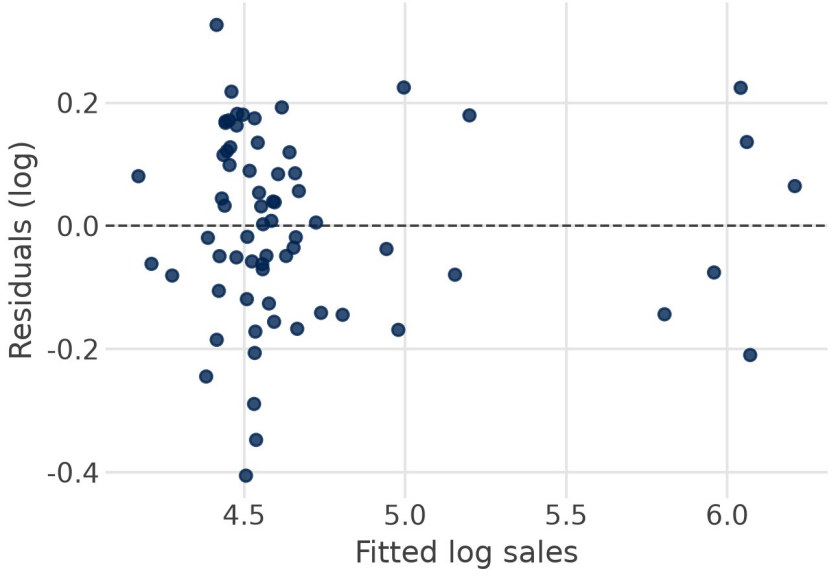
1. **H0 = the assumption holds:** constant variance, normal residuals, the right form
2. **p below 0.05:** evidence against H0, so the assumption is violated: act
3. **p above 0.05:** no evidence of a problem; not a proof that all is well
4. **Always look as well:** small samples miss problems, very large samples flag tiny ones

⚠ **Mind the direction:** in a coefficient test a small p is good news (an effect); in an assumption test it is bad news.

1 Linear form: look for a curve



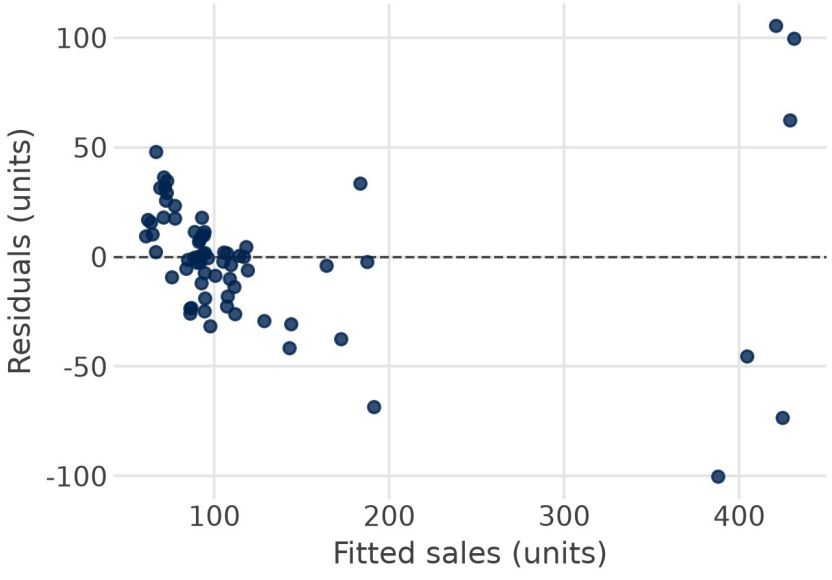
Linear model: a U-shape. RESET p < 0.001.



Log-log model: no curve. RESET p = 0.32.

RESET: adds the squared fitted values; if they help, the form is wrong.

2 Constant variance: look for a funnel



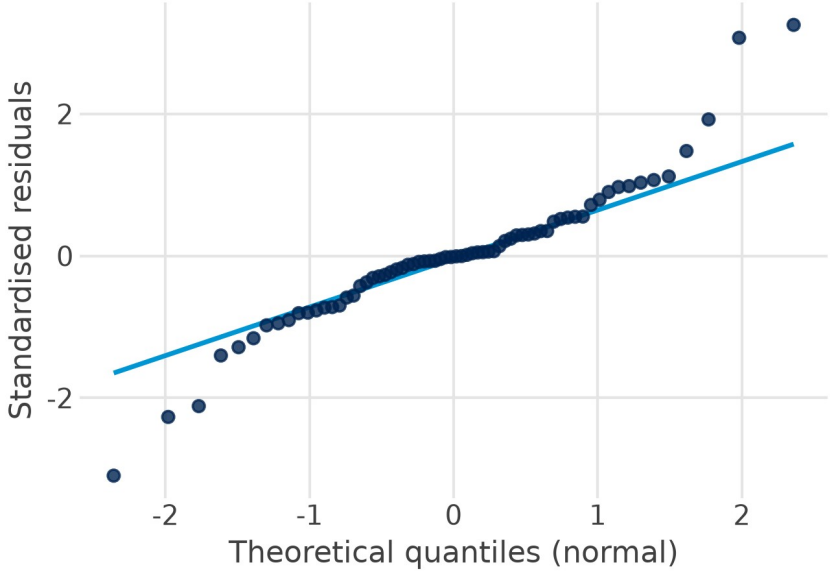
< 0.001 Breusch-Pagan p, linear model: unequal variance

0.68 Breusch-Pagan p, log-log model: no evidence

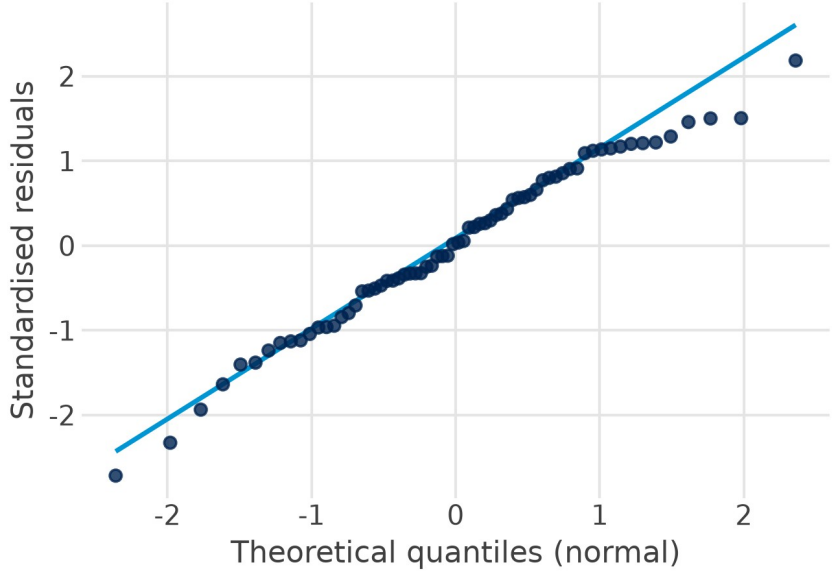
110 se of price1 in the linear model with HC3 errors (usual: 44)

Fix: logs, because sales vary in per cent; or robust errors with fit(cov_type="HC3").

3 Normal residuals: the Q-Q plot



Linear: heavy tails. Jarque-Bera $p < 0.001$.



Log-log: on the line. Jarque-Bera $p = 0.50$.

Matters least: with many observations the tests still work; extreme weeks matter more.

4 Independent errors: weeks follow weeks

≈ 2	Durbin-Watson without autocorrelation (range 0 to 4)
1.13	log-log model: positive autocorrelation between weeks
0.31	se of the elasticity with HAC errors, 1 lag (usual: 0.27)

Theory only today: Session 4 (time series) models autocorrelation with lags. Not in the exercise or the homework.

5 Multicollinearity: the VIF

$$VIF_j = 1 / (1 - R^2_j)$$

R^2_j: variable j regressed on all others

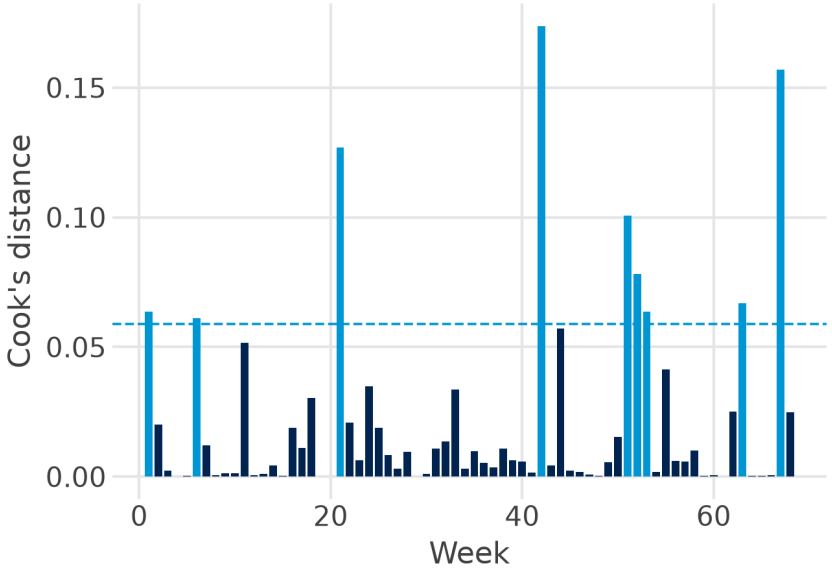
Above 5: warning; above 10: problem

Effect: large standard errors, unstable signs

Log-log model	VIF
ln price1	2.3
feature1	2.4
display1	1.1
… plus fand1:	
display1	7.6
fand1	7.8

Only three display weeks, mostly together with features: their effects cannot be separated.

6 Influential weeks: Cook's distance



Flag: 4/n = 0.059; nine weeks above

Week 42: 0.17; display 70 %, feature 42 %

Without it: elasticity −3.76 (−3.68)

But: display effect 0.45 instead of 0.61

Never delete silently: check the data, report with and without the week.

The scorecard: logs fix three assumptions at once

Test	H0	Linear	Log-log
RESET p	right form	< 0.001 ✗	0.32 ✓
Breusch-Pagan p	constant variance	< 0.001 ✗	0.68 ✓
Jarque-Bera p	normal residuals	< 0.001 ✗	0.50 ✓
Shapiro-Wilk p	normal residuals	0.001 ✗	0.50 ✓
Largest VIF	below 5	2.3 ✓	2.4 ✓
Durbin-Watson	about 2	1.48 ✗	1.13 ✗

Report the log-log model: sales respond in per cent, so errors grow with the level; logs fix that.

Dummy coding

Part 2 of 6 · 0/1 variables, categories, the reference group, the dummy trap

A dummy: 1 if yes, 0 if no

Week	december	sales
1	1	66
2	0	86
…	…	…
53	1	70

Linear: December −25 units, p = 0.32

Log-log: b = −0.319, p = 0.02

In per cent: e^b − 1 = e^−0.319 − 1 = −27 %

Reading a dummy: the difference to the weeks with 0, other variables held constant.

Small b (|b| < 0.1): 100 × b is close enough; larger b: use e^b − 1

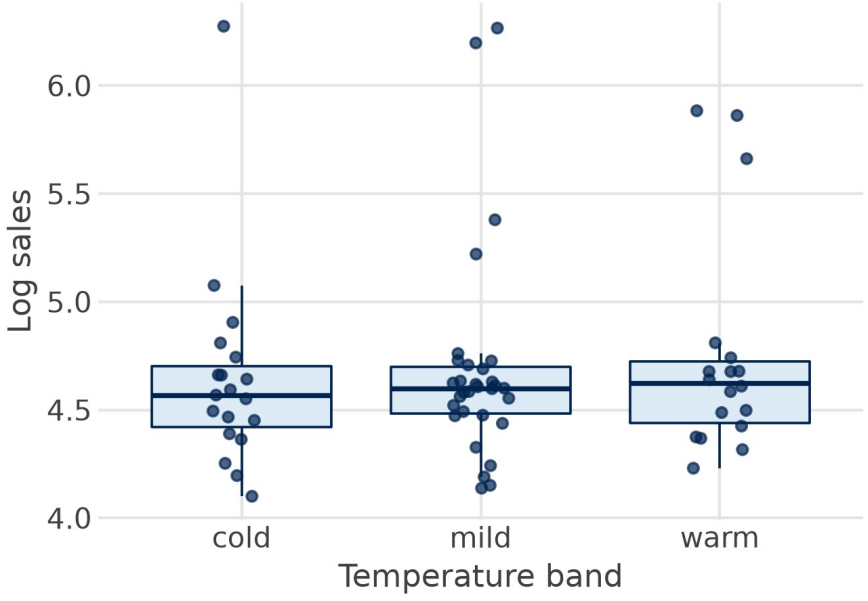
A category with k groups needs k − 1 dummies

Band	Weeks	cold	warm
cold, < 5 °C	19	1	0
mild, 5 to 14 °C	31	0	0
warm, ≥ 15 °C	18	0	1

mild = reference: the row of zeros

cold vs mild: +0.4 %, p = 0.94

warm vs mild: −7.1 %, p = 0.18



Result: with price and December in the model, the season adds little (F-test p = 0.33).

The dummy trap and the choice of reference

The dummy trap

cold + mild + warm = 1 in every week
= the intercept column: perfect collinearity
Always leave one group out

The reference group

Coefficients: differences to it
Changes the reading, not the fit (R^2 0.900)
Pick the normal case: mild

statsmodels formula

```
C(season)                          # reference: cold, first in the alphabet
C(season, Treatment("mild"))       # reference: mild
```

statsmodels avoids the trap when you write C(); with hand-made dummies, leave one out yourself.

Three pitfalls with dummies

One Easter week

- 1 in one week only
- fits it exactly (leverage 1.00)
- no real Easter effect

3 display weeks

- b = 0.61 rests on 3 weeks
- without week 42: 0.45
- count the 1s first

Level, not slope

- a dummy shifts the line
- a different slope needs
- an interaction

Before you trust a dummy: count how many 1s it has, e.g. (choc["display1"] > 0).sum().

Log-log regression

Part 3 of 6 · why logs, four log forms, reading and comparing them

Why logs: sales respond in per cent

```
sales    = a · price1^b · e^(c·december) · error          (multiplicative)
ln sales = ln a + b · ln price1 + c · december + ln error  (linear in logs)
```

Elasticities

b: per cent change in sales for 1 % more price

Comparable

across brands, markets and countries

Better errors

constant in per cent: the funnel disappears

No log of zero: shares with zeros (feature1 in 22 weeks) and dummies stay in levels.

Sources: Skiera, Reiner & Albers 2022; Hanssens, Parsons & Schultz 2001

Four forms, four readings of b

Form	Equation	b means	Brand 1
Linear	sales = a + b·price	+1 EUR: b units	−669: 10 cents, −67 units
Log-log	ln sales = a + b·ln price	+1 %: b % (elasticity)	−3.94: 1 %, −3.9 %
Log-linear	ln sales = a + b·price	+1: (e^b − 1)·100 %	−3.10: 10 cents, −27 %
Linear-log	sales = a + b·ln price	+1 %: b/100 units	−853: 1 %, −8.5 units

Log on the left: per cent change in sales. Log on the right: per cent change in the driver.

Sources: Wooldridge 2025, ch. 6; Skiera, Reiner & Albers 2022

Comparing a linear and a log model

Not R^2 or AIC: sales and log sales are different outcomes (0.884 vs 0.835 says nothing)

Residuals and tests: curve, funnel, Q-Q plot, RESET, Breusch-Pagan (Part 1)

Plausibility: sign and size of the coefficients, benchmarks such as the mean elasticity −2.6

Back on the sales scale: exp(fitted) for the log model, then compare predictions in units

Brand 1: the log-log model passes the tests, the linear one fails three. Report the log-log model.

Sources: Wooldridge 2025, ch. 6; Bijmolt, van Heerde & Pieters 2005

Non-linear relationships

Part 4 of 6 · curves in a linear regression:
squares, logs, the turning point

Linear in b, not in x

Diminishing

each step adds less

logs; square root

U or inverted U

an optimum or minimum

x and x^2

Thresholds

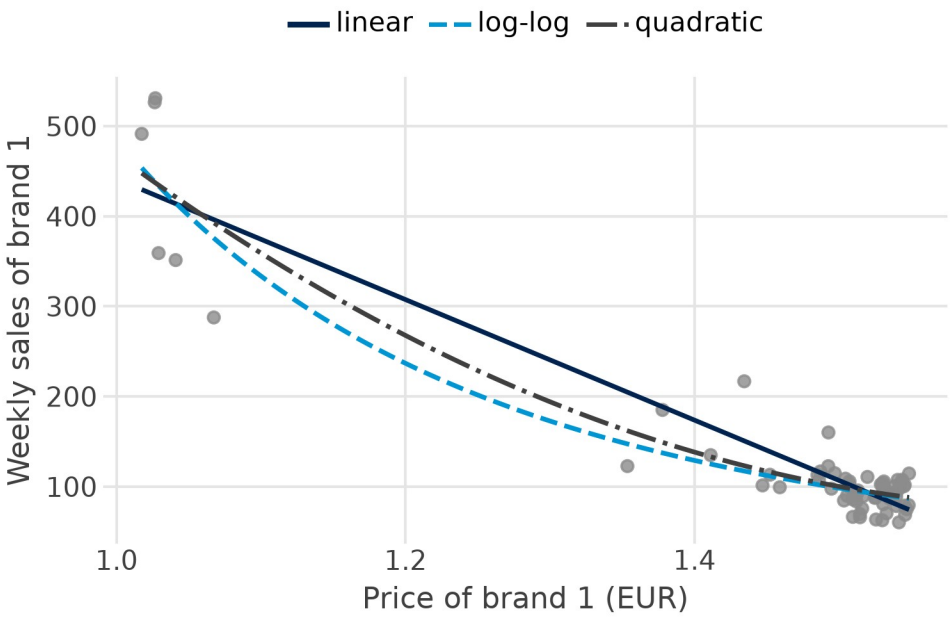
effect only above a level

dummies for ranges

```
sales = b0 + b1·price1 + b2·price1²        still linear in b0, b1, b2
smf.ols("sales ~ price1 + I(price1**2)", data=df)
```

Session 2: the marketing mix model adds saturation curves for media: the same idea.

Three curves through the same weeks



Linear: R^2 0.884; RESET p = 0.002

Quadratic: R^2 0.900; RESET p < 0.001

Log-log: R^2 0.835 (log sales); RESET p = 0.37

Thin middle: few weeks between 1.10 and 1.35

Look: all three agree where the data are; they differ where weeks are few.

The quadratic: the effect depends on the price

$$\text{sales} = 2{,}533 - 2{,}952 \cdot \text{price1} + 886 \cdot \text{price1}^2 \qquad \text{slope} = b1 + 2 \cdot b2 \cdot \text{price1}$$

Price (EUR)	Slope per EUR	10 cents more
1.00	−1,179	−118 units
1.25	−736	−74 units
1.50	−293	−29 units

1.67 turning point −b1 / (2·b2), in EUR: outside the data (1.02 to 1.55)

⚠ **Never read a curve beyond the data:** above 1.67 the quadratic says sales rise with price.

Still: RESET and Breusch-Pagan fail; the log-log model remains the better form

Choosing the form: four questions

1. **Theory:** saturation, an optimum, effects in per cent?
2. **Look and test:** residuals against fitted, RESET
3. **Compare like with like:** same outcome: adjusted R^2 or AIC; else tests and predictions
4. **Plausible in the data range:** signs, turning points, no extrapolation

Example: temp² in the full log-log model, p = 0.14: keep temperature linear.

The analyses in Python

Part 5 of 6 · packages, two models, plots, tests, dummies, curves

The packages and functions

Task	Package	Function
Fit the models	statsmodels	smf.ols(formula, data).fit()
Curve: RESET	statsmodels	linear_reset(model, power=2, use_f=True)
Funnel: Breusch-Pagan	statsmodels	het_breuschpagan(resid, exog)
Normality	statsmodels, scipy	jarque_bera(resid); stats.shapiro(resid)
Multicollinearity	statsmodels	variance_inflation_factor(exog, i)
Influence	statsmodels	model.get_influence().cooks_distance
Dummies	polars, statsmodels	pl.when().then(); C(season, Treatment("mild"))
Plots	plotnine	geom_point, geom_qq, geom_qq_line

Step 1: fit both models

Python: polars and statsmodels

```
import numpy as np
import polars as pl
import statsmodels.formula.api as smf

choc = pl.read_csv(
    "data/legacy/chocolate_dataset.csv")
df = choc.to_pandas()

lin = smf.ols("sales ~ price1 + feature1 + display1"
              " + temp + december", data=df).fit()
ll = smf.ols("np.log(sales) ~ np.log(price1)"
             " + np.log(price2) + np.log(price3)"
             " + np.log(price4) + feature1 + display1"
             " + temp + december", data=df).fit()
```

	Linear	Log-log
Own price	−678	−3.68
R^2	0.896	0.903
n	68	68

Keep both: every next step uses lin and ll.

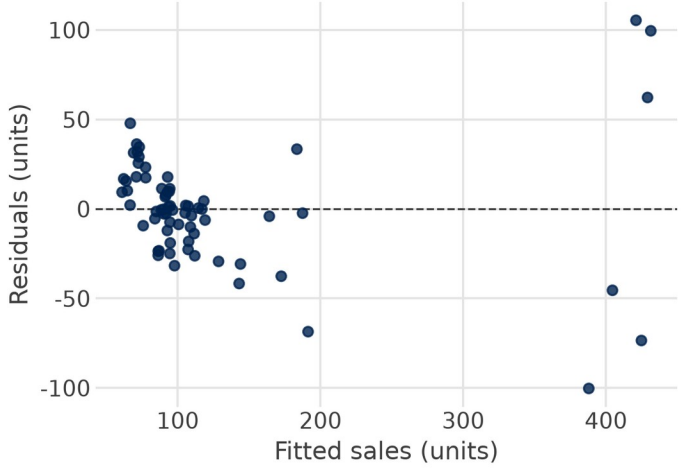
Step 2: look at the residuals

Python: plotnine

```
from plotnine import (aes, geom_hline, geom_point,
    geom_qq, geom_qq_line, ggplot, labs)

df["fit"], df["res"] = lin.fittedvalues, lin.resid
(ggplot(df, aes("fit", "res"))
 + geom_point()
 + geom_hline(yintercept=0, linetype="dashed")
 + labs(x="Fitted sales", y="Residuals"))

df["z"] = (df["res"] - df["res"].mean()) / df["res"].std()
(ggplot(df, aes(sample="z"))
 + geom_qq() + geom_qq_line()
 + labs(x="Normal quantiles", y="Residuals"))
```



lin: curve and funnel. Then the same for ll.

Run it twice: for lin and for ll; compare curve, funnel and the Q-Q line.

Step 3: test the assumptions

Python: statsmodels and scipy

```
from scipy import stats
from statsmodels.stats.diagnostic import (
    het_breuschpagan, linear_reset)
from statsmodels.stats.stattools import jarque_bera
from statsmodels.stats.outliers_influence import (
    variance_inflation_factor)

for m in (lin, ll):
    X = m.model.exog
    print(linear_reset(m, power=2, use_f=True).pvalue,
          het_breuschpagan(m.resid, X)[1],
          jarque_bera(m.resid)[1],
          stats.shapiro(m.resid).pvalue,
          max(variance_inflation_factor(X, i)
              for i in range(1, X.shape[1])))
```

p / VIF	lin	ll
RESET	< .001	0.32
BP	< .001	0.68
JB	< .001	0.50
SW	0.001	0.50
max VIF	2.3	2.4

Robust errors if needed: lin.model.fit(cov_type="HC3").

Step 4: dummies from a category

Python: polars and statsmodels

```
choc = choc.with_columns(
    pl.when(pl.col("temp") < 5).then(pl.lit("cold"))
    .when(pl.col("temp") < 15).then(pl.lit("mild"))
    .otherwise(pl.lit("warm")).alias("season"))
choc["season"].value_counts()      # 19, 31, 18
df = choc.to_pandas()

sea = smf.ols("np.log(sales) ~ np.log(price1)"
              " + np.log(price2) + np.log(price3)"
              " + np.log(price4) + feature1 + display1"
              " + december"
              " + C(season, Treatment('mild'))",
              data=df).fit()
np.exp(sea.params) - 1             # per cent
```

vs mild	e^b − 1	p
cold	+0.4 %	0.94
warm	−7.1 %	0.18
december	−28.5 %	0.01

k − 1 dummies: C() leaves the reference out for you.

Step 5: a curve and its turning point

Python: statsmodels

```
quad = smf.ols("sales ~ price1 + I(price1**2)",
               data=df).fit()
b1 = quad.params["price1"]
b2 = quad.params["I(price1 ** 2)"]
turn = -b1 / (2 * b2)                # 1.67
choc["price1"].min(), choc["price1"].max()

for p in (1.00, 1.25, 1.50):         # slope at p
    print(p, (b1 + 2 * b2 * p) / 10)  # per 10 cents

linear_reset(quad, power=2, use_f=True).pvalue
```

Result	Value
b1	−2,952
b2	886
Turning point	1.67 EUR
Price range	1.02 to 1.55
RESET p	< 0.001

Outside the data: the turning point says nothing about real prices.

In-class exercise

Part 6 of 6 · online quiz, then the chocolate data in Python, then with Posit Assistant

Online quiz 3: assumptions, dummies, logs in five questions

1 Breusch-Pagan gives p = 0.001. This means …
A constant variance B unequal variance (heteroscedasticity) C normal residuals

2 Four store formats in one model with an intercept need …
A 4 dummies B 3 dummies C 1 dummy

3 In a log model the December dummy is b = −0.36. December weeks sell …
A 0.36 units fewer B 36 % fewer C about 30 % fewer

4 A quadratic in price turns at 1.67 EUR; prices in the data run 1.02 to 1.55. So …
A do not read it: outside the data B raise the price above 1.67 C sales rise with price

5 Linear model R^2 0.88, log-log R^2 0.84. Which is better?
A the linear one B R^2 cannot tell: different outcomes C the log-log one

Online on Canvas: 3 minutes alone, then compare with your neighbour.

Your turn: which model may brand 1 trust?

Pairs, with packages: 15 minutes	Plenary, answers: 5 minutes	Posit Assistant: 10 minutes

1. **Look:** sales ~ price1 and np.log(sales) ~ np.log(price1); residual plot of each: curve, funnel?
2. **Test:** RESET, Breusch-Pagan and Jarque-Bera for both models; which fails what?
3. **Dummies:** season from temp with polars; add december and C(season, Treatment("mild")); read both in per cent
4. **Curve:** sales ~ price1 + I(price1**2); turning point; inside the price range?
5. **The manager:** which model do you report, and one sentence with a number

File: `sessions/01-foundations/assumptions_exercise.qmd` (data, tasks, Posit Assistant)

Then: the same tasks with Posit Assistant

1. **Code first:** solve the tasks with the packages; keep the cells run, so Posit Assistant sees choc, df and the models
2. **Open the chat:** Posit Assistant icon in the activity bar, or Ctrl/Cmd + Shift + P → View: Show Posit Assistant
3. **Give context:** type @assumptions_exercise.qmd, then one task per prompt: the file, the packages, the output
4. **Ask for meaning:** let it explain each test result in one sentence; check the direction of p
5. **Compare:** its code, numbers and wording with yours and the answers

Stay in Normal mode: never /yolo. Read every edit before you accept it. You own every number and every sentence.

Prompts for the assumption checks

Prompt for task 2

```
@assumptions_exercise.qmd Task 2.
Run RESET, Breusch-Pagan and
Jarque-Bera with statsmodels for
lin1 and ll1. Show the p-values
in one table and explain in one
sentence each which assumption
fails.
```

Prompt for task 3

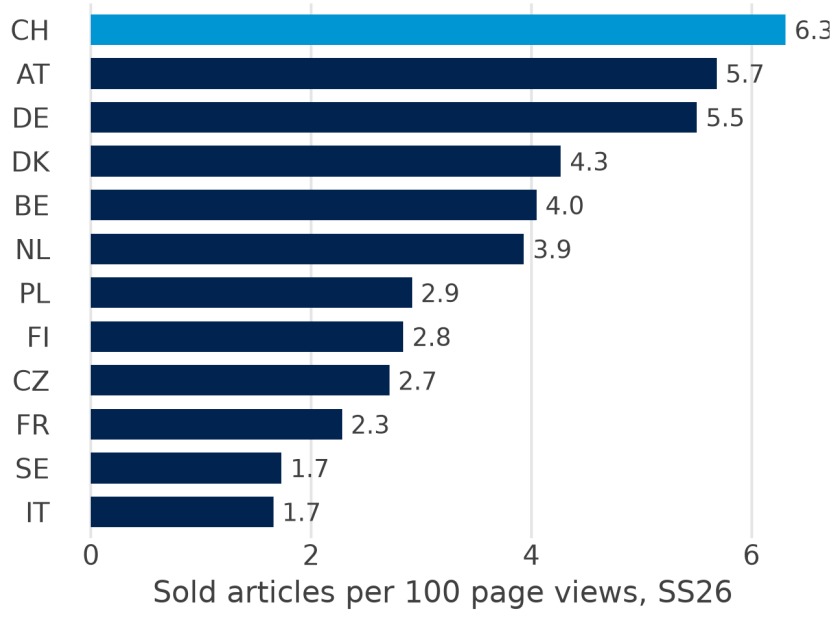
```
Task 3. Build season from temp
with polars when/then (cold < 5,
mild < 15, warm). Fit the model
with C(season, Treatment("mild"))
and give each dummy in per cent
with exp(b) - 1.
```

Name three things: the file, the packages (polars, statsmodels, scipy), the output

Check the direction: in an assumption test a small p means a problem

Watch for slips: tests on sales instead of residuals, three dummies plus intercept, 100 × b

Second case: one fashion brand, twelve countries



Data: 403 rows, season × country × article type

Funnel: page views → add to basket → sold → returns

Markets: DE, CH, AT = 71 % of NMV in SS26

New in SS26: Switzerland converts best at once

The manager asks: where do page views turn into sales, and should we invest in Switzerland?

Your turn: the fashion brand in Python

Pairs, with packages: 20 minutes	Plenary, answers: 5 minutes	Posit Assistant: 10 minutes

1. **Markets:** SS26 by country: NMV, sales, sold per 100 views, return rate; a great_tables table
2. **Funnel:** np.log(sold_articles) ~ np.log(pdp_views), rows with sales only; read the elasticity
3. **Country dummies:** + C(country, Treatment("DE")) + C(season); per cent effects; two reasons
4. **Price, discount, sizes:** add them; is the price coefficient a price elasticity? Check with HC3
5. **Switzerland:** CH vs DE and AT; why is nmv / (sold × price) higher? Invest or not?

File: `sessions/01-foundations/fashion_exercise.qmd` (data: data/fashion/fashion_countries.csv)

Key takeaways

1. **Look, then test, then fix:** residual and Q-Q plots; RESET, Breusch-Pagan, Jarque-Bera, VIF, Cook's distance

2. **In assumption tests** a small p is bad news

3. **Dummies:** k − 1 for k groups; each b is a difference to the reference; count the 1s

4. **Logs:** elasticities, per cent effects, better residuals; compare models by tests, not R^2

5. **Curves:** squares and logs keep OLS linear in b; never read a curve beyond the data

Homework: coding exercise 2 with your personal data, brief on Canvas.

References

Academic

Skiera, B., Reiner, J. & Albers, S. (2022). Regression analysis. In Homburg et al. (eds), Handbook of Market Research, 299–327. Springer.

Bojinov, I. I., Parzen, M. & Hamilton, P. J. (2025). Linear Regression. HBS note 9-622-100 (Canvas).

Wooldridge, J. M. (2025). Introductory Econometrics, 8th ed. Cengage. Ch. 6 to 9.

Hanssens, D. M., Parsons, L. J. & Schultz, R. L. (2001). Market Response Models. Kluwer.

Hair, J. F. et al. (2019). Multivariate Data Analysis, 8th ed. Cengage. Ch. 5.

Learning resources

James, G. et al. (2023). ISLP, ch. 3.3 and the Python lab (free).

statsmodels. Regression diagnostics and specification tests (documentation).

Gelman, Hill & Vehtari (2020). Regression and Other Stories, ch. 11 to 12 (free PDF).

Turrell, A. Coding for Economists: regression diagnostics (free).

polars. when/then/otherwise and to_dummies (documentation).

VIENNA UNIVERSITY OF ECONOMICS AND BUSINESS

INSTITUTE FOR INTERNATIONAL BUSINESS
Welthandelsplatz 1, 1020 Vienna, Austria

DR ARNE FLOH
Senior Lecturer in Marketing

F +43-31336-6367
arne.floh@wu.ac.at
www.wu.ac.at/en/welthandel/arne-floh
Office hours: book online